

TALAT ILYASS

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PROFILE

B.Sc. Biochemistry & Bioinformatics researcher combining wet-lab molecular biology with *in silico* structural analysis. Proficient in executing core biochemical assays (PCR, Western blotting, DNA extraction) and independently optimizing lab workflows through custom Python scripting. Technically fluent in terminal-based environments (Bash, Zsh) and molecular visualization tools (PyMOL, VMD, UCSF ChimeraX). Experience applying this wet-dry toolkit to sequence-level and structural analysis of kinase signaling cascades.

EDUCATION

Bachelor of Science (B.Sc.) in Biochemistry [2023 – 2026]

S.P. College, Srinagar — Cluster University of Srinagar

Relevant coursework: Biochemistry, Molecular Biology, Microbiology, Bioinformatics, Biostatistics.

RESEARCH EXPERIENCE

Research Intern – S.P. College, Srinagar [January 2026] – [March 2026]

Department of Biochemistry

Supervisor: Dr. Shafat A.

Project: Molecular crosstalk between mTOR and p70S6 kinase in actin cytoskeletal dynamics

- Investigated how p70S6K modulates actin cytoskeleton remodeling through phosphorylation-dependent signaling pathways, with a focus on structural domains of actin-interacting proteins.
- Synthesized a comprehensive scientific review literature; produced presentation decks for professorial review and mentorship feedback.

PROJECTS

Codon Usage Bias Analysis: Prefrontal Cortex vs. Liver

Independent Computational Biology Project [\[link\]](#)

- Built an end-to-end pipeline (Python, Biopython, NCBI Entrez API) comparing codon usage bias (CAI, RSCU) across 28 human genes highly expressed in prefrontal cortex vs. Liver.
- Constructed a CAI reference dataset from scratch using 10 human ribosomal protein genes and independently caught and corrected multiple pipeline errors by manually validating results against an established tool before trusting automated output at scale.
- Built an interactive data visualization dashboard and published the full pipeline and write-up on GitHub.

TRAINING

Mentee – SPROUT Program, Summer Cohort 2026 [Jun 2026 – Present]

JK Scientists, (Support Program for Research Outreach and Upskill Technology), J&K, India

- Received mentorship from researchers at national and international institutions on academic planning, CV writing, and SOP preparation.
- Participated in structured sessions on profile building, research methodology, international internship/scholarship strategy, AI & ML career pathways, and innovation ecosystems.

Certifications

Sri Pratap college, Srinagar

Fine-Tuning LLM's

Invisible technologies

SKILLS

- **Wet-Lab:** DNA extraction, PCR, agarose gel electrophoresis, colorimetric assays, standard-curve analysis, competent cell preparation, and basic microbiology techniques.
- **Computational:** Python (pandas, NumPy), Linux/terminal (bash, zsh), Git, Jupyter/Colab, LaTeX, Excel/Sheets, PyMOL, BLAST, NCBI/UniProt, sequence alignment & structural visualization, UCSC Genome Browser, PubMed.

Websites/ Press

Website: talatilyass.app

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REFEREES

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